Grade: Check Plus

ATM Withdrawal Distribution

x ($)	f(x)	μ	σ
10	0.10	1.00	160.00
20	0.15	3.00	135.00
40	0.30	12.00	30.00
60	0.15	9.00	15.00
80	0.25	20.00	225.00
100	0.05	5.00	125.00
Total	**1.00**	**50.00**	**690.00**
		μ = $50	**σ = $26.27**

a) P(X ≥ $60)

all values at or above $60: ✓ Qa

P(X ≥ 60) = f(60) + f(80) + f(100)

= 0.15 + 0.25 + 0.05

*** P(X ≥ $60) = 0.45**

b) E(X) = Σ x · f(x) = 1 + 3 + 12 + 9 + 20 + 5 ✓ Qb

*** μ = $50.00**

Var(X) = E(X²) − μ² = 3,190 − 2,500 = 690

σ = √690

*** σ = $26.27**

✓ Qc

c) This represents the long-run average over many customers,

not a value any single customer must withdraw so the manager is wrong.

d)

μ ± σ = 50 ± 26.27 = $76.27 ✓ Qd

X = 40 and X = 60

P = f(40) + f(60) = 0.30 + 0.15 **= 0.45**

e) μ + 2σ = 102.54, μ - 2σ = −$2.54 ✓ Qe

All values {10, 20, 40, 60, 80, 100} fall within this range.